

Auteur: Sabawoon Enayat
 Studierichting: Informatica
 Oefeningenreeks 4A nr. 41.12

$$\text{Opgave: } I = \int \frac{dx}{\sqrt{x^2 + 5x + 8}}$$

$$(x - a)^2 = x^2 + 5x + b \Rightarrow D = 5^2 - 4 \cdot 1 \cdot b = 25 - 4b = 0$$

$$\Rightarrow b = \frac{25}{4} \stackrel{4 \cdot 8 = 32}{\Rightarrow} 8 = \frac{25}{4} + \frac{7}{4}$$

$$I = \int \frac{dx}{\sqrt{\left(x^2 + 5x + \frac{25}{4}\right) + \frac{7}{4}}}$$

$$\Rightarrow D = 0 \Rightarrow x = \frac{-5}{2}$$

$$I = \int \frac{dx}{\sqrt{\left(x + \frac{5}{2}\right)^2 + \frac{7}{4}}}$$

$$I = \int \frac{d\left(x + \frac{5}{2}\right)}{\sqrt{\left(x + \frac{5}{2}\right)^2 + \frac{7}{4}}}$$

$$= \ln \left| x + \frac{5}{2} + \sqrt{\left(x + \frac{5}{2}\right)^2 + \frac{7}{4}} \right| + c$$

References